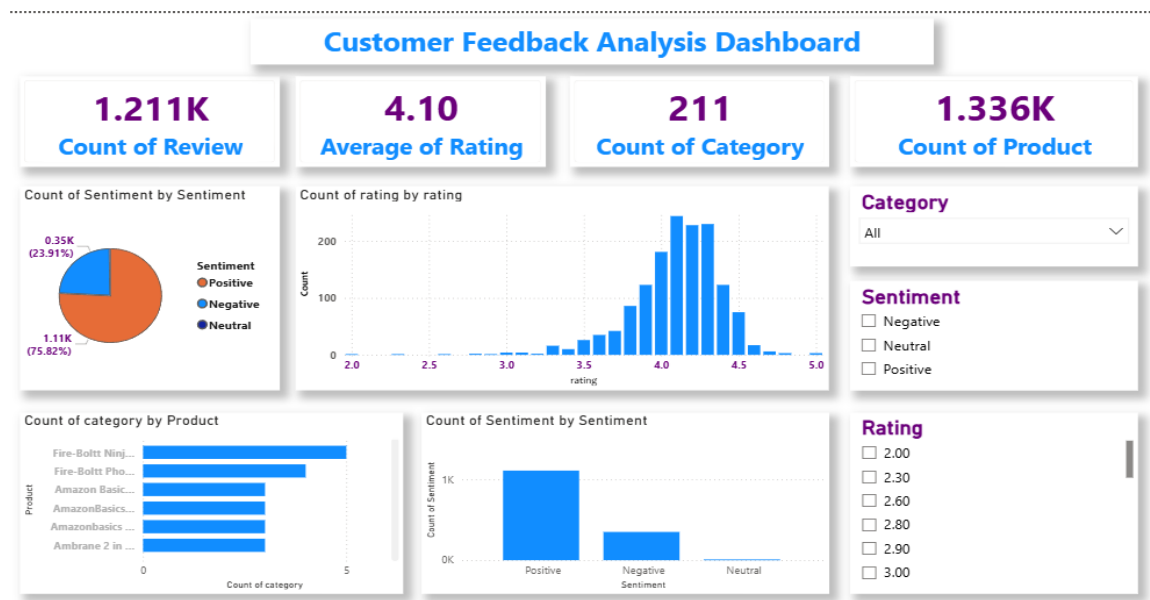


Data-Driven Storytelling and Visualization Documentation

Project Statement 11: Customer Feedback Analysis



Submitted by: Aditya Kumar Gupta, Kanchana Dongre, Jigyasa Patle, Adarsh Kumar

Submitted To: L&T EduTech

University: SAM Global University

Branch: B.Tech CSE

Batch No: 11

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1. Introduction

Customer Feedback Analysis is an important process used to understand customer opinions, satisfaction levels, and product performance through customer reviews and ratings. Businesses use customer feedback analysis to improve product quality, customer service, and overall user experience.

In this project, customer review data was analyzed using Python and Power BI. Different data analysis and visualization techniques were applied to identify customer sentiments, rating patterns, and feedback trends. Natural Language Processing (NLP) techniques such as text preprocessing, sentiment analysis, word cloud generation, and topic modeling were used to extract meaningful insights from customer reviews.

The project also included the development of an interactive Power BI dashboard to visualize customer sentiments, product categories, ratings, and feedback analysis. The dashboard helps users easily explore and understand customer review patterns through charts, filters, and visual reports.

This project demonstrates the practical implementation of data analytics, NLP, and business

intelligence tools for customer feedback analysis and decision-making.

Screenshot:

df.head()

Python

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product	
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories Accessories&Peripherals ...	₹399	₹1,099	64%	4.2	24,269	High Compatibility : Compatible With iPhone 12...	AG3D6O4STAQKAY2UVG
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories Accessories&Peripherals ...	₹99	₹349	43%	4.0	43,994	Compatible with all Type C enabled devices, be...	AECPFYFQVRUWC3KC
2	B096MSW6CT	Source Fast Phone Charging Cable & Data Sync U...	Computers&Accessories Accessories&Peripherals ...	₹199	₹1,899	90%	3.9	7,928	【 Fast Charger& Data Sync】 -With built-in safet...	AGU3BBQ2V2DDAMOAK
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories Accessories&Peripherals ...	₹329	₹699	53%	4.2	94,363	The boAt Deuce USB 300 2 in 1 cable is compati...	AEWAZDZZJLQUYVOVK
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	Computers&Accessories Accessories&Peripherals ...	₹154	₹399	61%	4.2	16,905	[CHARGE & SYNC FUNCTION]- This cable comes wit...	AE3Q6KSUK5P75D5HF

df.info()

<class 'pandas.DataFrame'>
RangeIndex: 1465 entries, 0 to 1464
Data columns (total 16 columns):
Column Non-Null Count Dtype
--- --
0 product_id 1465 non-null str
1 product_name 1465 non-null str
2 category 1465 non-null str
3 discounted_price 1465 non-null str
4 actual_price 1465 non-null str
5 discount_percentage 1465 non-null str
6 rating 1465 non-null str
7 rating_count 1463 non-null str
8 about_product 1465 non-null str
9 user_id 1465 non-null str
10 user_name 1465 non-null str
11 review_id 1465 non-null str
12 review_title 1465 non-null str
13 review_content 1465 non-null str
14 img_link 1465 non-null str
15 product_link 1465 non-null str
dtypes: str(16)
memory usage: 183.3 KB

2. Objective

The main objectives of this project are:

- Analyze customer reviews and ratings
 - Identify customer sentiments
 - Extract common feedback keywords
 - Perform topic modeling analysis
 - Create visualizations and dashboard reports
 - Build an interactive Power BI dashboard
 - Understand customer satisfaction trends
-

3. Dataset Information

The dataset used in this project was collected from Kaggle and contains Amazon product review and customer feedback data. The dataset includes product-related information such as product names, categories, prices, ratings, discount percentages, customer review titles, and review content.

The dataset was used to analyze customer sentiments, product ratings, customer opinions, and feedback trends. Python libraries and Power BI were used for preprocessing, sentiment analysis, visualization, and dashboard development.

The dataset contains both structured and unstructured data. Structured data includes ratings, prices, and categories, while unstructured data includes customer review text used for Natural Language Processing (NLP) tasks such as sentiment analysis, word cloud generation, and topic modeling.

Before analysis, the dataset was cleaned and preprocessed by handling missing values, converting rating columns into numeric format, and cleaning review text using NLP preprocessing techniques.

Dataset Features

- product_id – Unique product identifier
- product_name – Name of the product
- category – Product category
- discounted_price – Discounted product price
- actual_price – Original product price
- discount_percentage – Discount percentage on product
- rating – Customer rating of the product
- rating_count – Total number of ratings
- about_product – Product description
- review_title – Title of customer review
- review_content – Customer review text
- user_id – Unique user identifier
- user_name – Customer name

- review_id – Unique review identifier
- img_link – Product image link
- product_link – Amazon product link

Dataset Analysis

The dataset was analyzed using data analysis and visualization techniques to identify customer feedback patterns, product performance, and sentiment trends.

The dataset was further used for:

- Sentiment Analysis
- Word Cloud Analysis
- Topic Modeling
- Product Rating Analysis
- Category-wise Analysis
- Dashboard Visualization
- Interactive Reporting

Screenshot:

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 1465 entries, 0 to 1464
Data columns (total 16 columns):
#   column                Non-Null Count  Dtype
---  -
0   product_id            1465 non-null   str
1   product_name          1465 non-null   str
2   category              1465 non-null   str
3   discounted_price      1465 non-null   str
4   actual_price           1465 non-null   str
5   discount_percentage   1465 non-null   str
6   rating                1465 non-null   str
7   rating_count          1463 non-null   str
8   about_product         1465 non-null   str
9   user_id               1465 non-null   str
10  user_name              1465 non-null   str
11  review_id             1465 non-null   str
12  review_title          1465 non-null   str
13  review_content        1465 non-null   str
14  img_link              1465 non-null   str
15  product_link          1465 non-null   str
dtypes: str(16)
memory usage: 183.3 KB
```

```
df.isnull().sum()
```

```
product_id      0
product_name    0
category        0
discounted_price 0
actual_price    0
discount_percentage 0
rating          0
rating_count    2
about_product   0
user_id         0
user_name       0
review_id       0
review_title    0
review_content  0
img_link        0
product_link    0
dtype: int64
```

4. Technologies Used

Technology Purpose

Python Data Analysis

Pandas Data Handling

NumPy Numerical Operations

Matplotlib Visualization

Seaborn Statistical Charts

NLTK Text Processing

Scikit-learn Topic Modeling

WordCloud Keyword Visualization

Power BI Dashboard Development

5. Data Preprocessing

Data preprocessing is one of the most important steps in data analysis. In this project, preprocessing was performed to clean and prepare the dataset for analysis.

The preprocessing steps included:

- Removing null values
- Converting ratings into numeric format
- Cleaning review text
- Removing punctuation and stopwords
- Tokenization and text normalization
- Creating sentiment labels

These preprocessing steps improved the quality and consistency of the dataset before performing sentiment analysis and visualization.

The review text was cleaned using Natural Language Processing techniques to remove unnecessary symbols, stopwords, and unwanted characters.

Screenshot:

```
df[['Review', 'Cleaned_Review']].head()
```

	Review	Cleaned_Review
0	Looks durable Charging is fine tooNo complains...	looks durable charging fine toono complains ch...
1	I ordered this cable to connect my phone to An...	ordered cable connect phone android auto car c...
2	Not quite durable and sturdy,https://m.media-a...	quite durable sturdy https media amazon com im...
3	Good product,long wire,Charges good,Nice,I bou...	good product long wire charges good nice bough...
4	Bought this instead of original apple, does th...	bought instead original apple work rs fast app...

6. Sentiment Analysis

Sentiment analysis was performed to classify customer reviews into Positive, Neutral, and Negative categories based on customer ratings.

The sentiment classification logic used was:

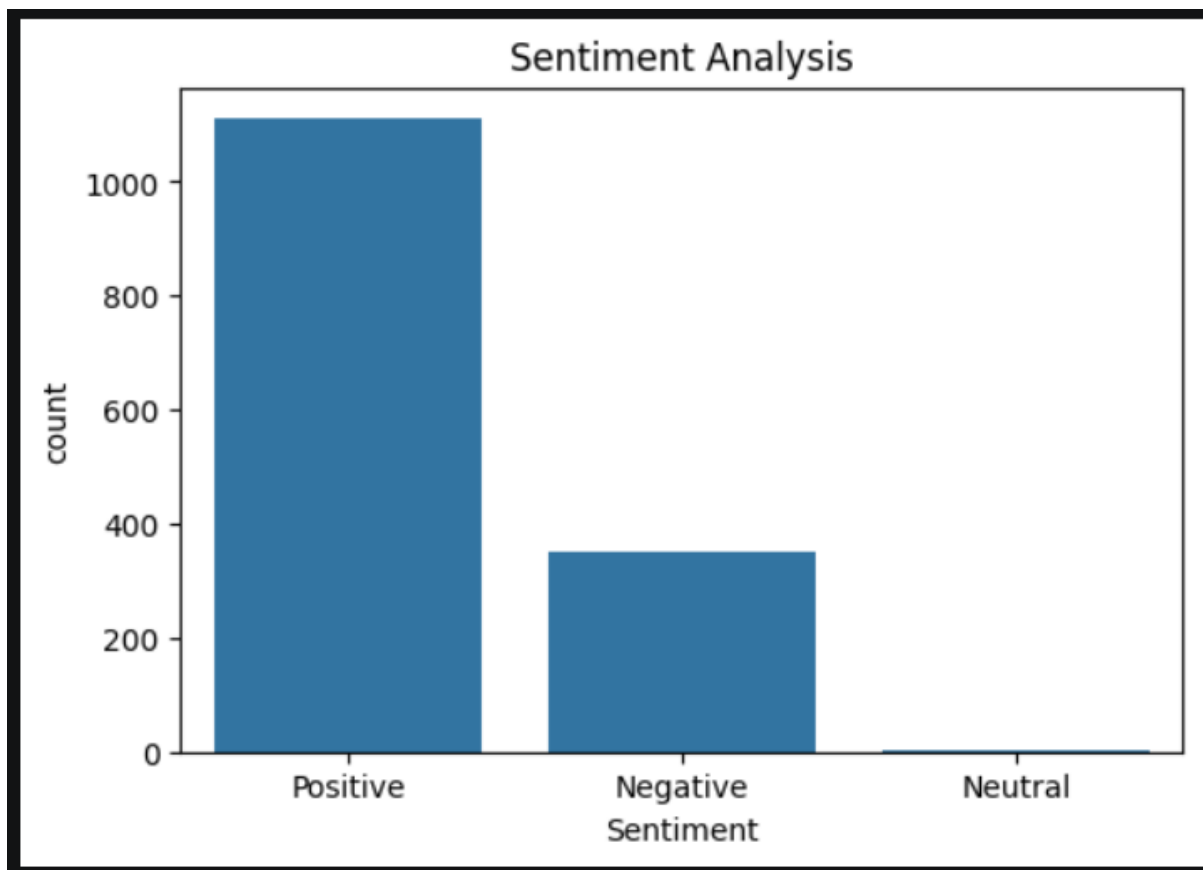
- Ratings greater than or equal to 4 → Positive
- Rating equal to 3 → Neutral
- Ratings below 3 → Negative

The analysis showed that most customer reviews were positive, indicating overall customer satisfaction with the products.

Different charts and visualizations were created to analyze sentiment distribution and customer opinion trends.

```
df['Sentiment'].value_counts()
```

```
Sentiment
Positive    1110
Negative     350
Neutral         4
Name: count, dtype: int64
```



7. Word Cloud Analysis

Word Cloud analysis was used to identify the most frequently occurring words in customer reviews. Frequently appearing words were displayed in larger size in the visualization.

This analysis helped identify common customer opinions, product-related discussions, and repeated feedback themes.

The word cloud provided quick visual insights into important keywords used by customers in their reviews.

Topic modeling grouped similar words together to identify customer concerns, product quality discussions, charging performance, cable durability, product quality, and customer experience.

The extracted topics provided meaningful insights into customer feedback trends and product performance.

```
cv = CountVectorizer(  
    max_df=0.95,  
    min_df=2,  
    stop_words='english'  
)  
  
dtm = cv.fit_transform(df['Cleaned_Review'])
```

```
lda = LatentDirichletAllocation(  
    n_components=5,  
    random_state=42  
)  
  
lda.fit(dtm)
```

▼ LatentDirichletAllocation ⓘ ?
► Parameters

```
for index, topic in enumerate(lda.components_):  
    print(f"\nTopic {index+1}")  
    print([cv.get_feature_names_out()[i]  
          for i in topic.argsort()[-10:]])
```

Topic 1
['like', 'value', 'working', 'money', 'use', 'price', 'nice', 'cable', 'quality', 'product']

Topic 2
['power', 'screen', 'price', 'time', 'like', 'watch', 'quality', 'camera', 'battery', 'phone']

Topic 3
['price', 'used', 'time', 'like', 'using', 'watch', 'water', 'easy', 'use', 'product']

Topic 4
['like', 'using', 'fast', 'price', 'product', 'sound', 'charging', 'tv', 'quality', 'cable']

Topic 5
['nice', 'sy', 'jpg', 'media', 'https', 'com', 'amazon', 'images', 'quality', 'product']

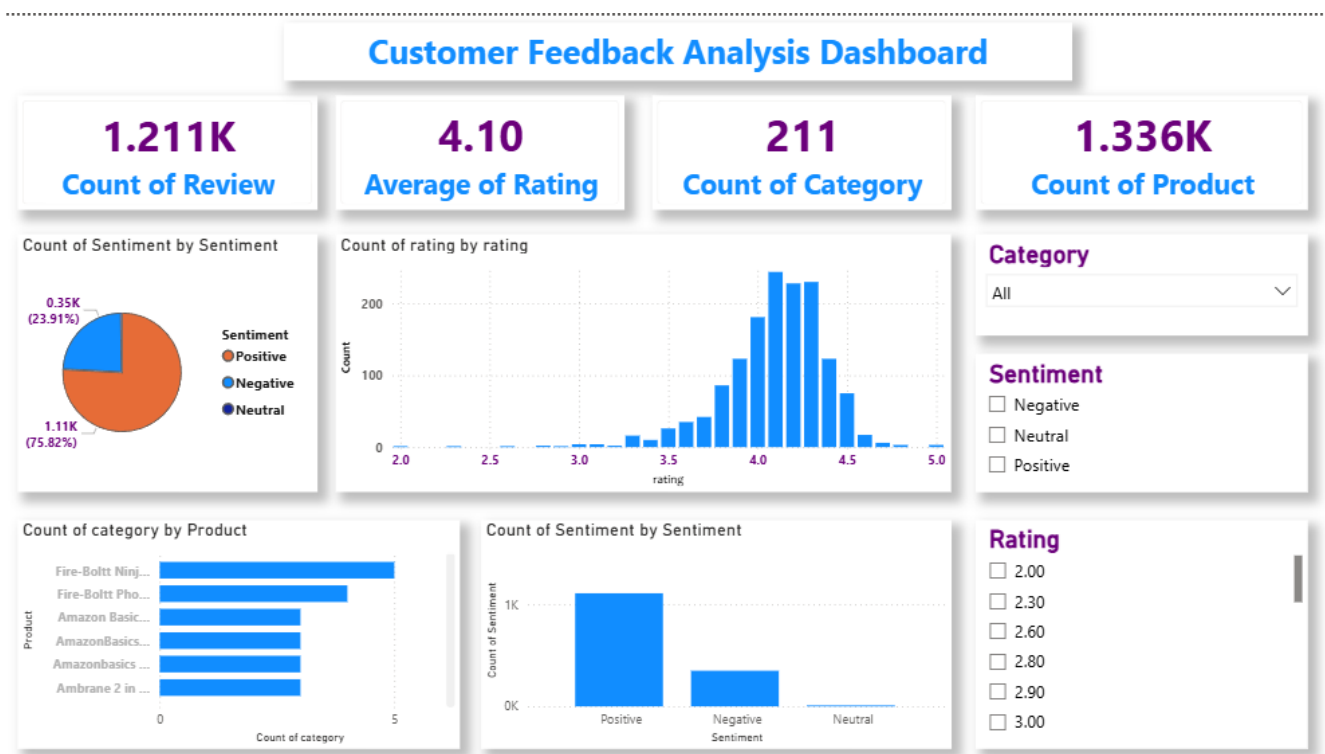
9. Power BI Dashboard

An interactive Power BI dashboard was created to visualize customer review analysis results. The dashboard included charts, graphs, filters, and tables for interactive exploration of customer feedback.

The dashboard features included:

- Sentiment Distribution Pie Chart
- Category-wise Product Analysis
- Rating Distribution Analysis
- Discount Percentage Analysis
- Product-wise Review Table
- Interactive Filters and Slicers

The dashboard helped users analyze customer sentiments, ratings, and product categories efficiently.



10. Conclusion

The project successfully analyzed Amazon customer feedback using Python and Power BI. Various preprocessing, sentiment analysis, visualization, and topic modeling techniques were applied to extract meaningful insights from customer reviews.

The analysis showed that most customers provided positive feedback, indicating overall customer satisfaction. Word cloud analysis and topic modeling helped identify common customer discussion themes and product-related concerns.

The interactive Power BI dashboard provided an effective way to visualize and explore customer feedback trends, sentiments, ratings, and category-wise product analysis.

This project demonstrated the practical implementation of data analytics, Natural Language Processing (NLP), and business intelligence tools in real-world customer feedback analysis.

Github Repo Link:

<https://github.com/maiadityahoonn/Customer-Feedback-Analysis>